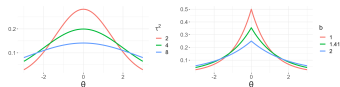
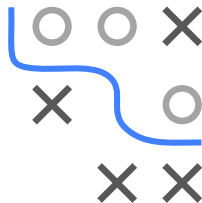


Introduction to Machine Learning

Regularization Bayesian Priors



Learning goals

- RRM is same as MAP in Bayes
- Gaussian/Laplace prior corresponds to $L2/L1$ penalty

RRM VS. BAYES

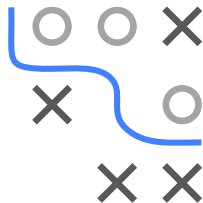
We already created a link between max. likelihood estimation and ERM.

Now we will generalize this for RRM.

Assume we have a parameterized distribution $p(y|\theta, \mathbf{x})$ for our data and a prior $q(\theta)$ over our param space, all in Bayesian framework.

From Bayes theorem:

$$p(\theta|\mathbf{x}, y) = \frac{p(y|\theta, \mathbf{x})q(\theta)}{p(y|\mathbf{x})} \propto p(y|\theta, \mathbf{x})q(\theta)$$



EXAMPLE: BAYESIAN L2 REGULARIZATION

We can easily see the equivalence of $L2$ regularization and a Gaussian prior:

- Gaussian prior $\mathcal{N}_d(\mathbf{0}, \text{diag}(\tau^2))$ with uncorrelated components for $\boldsymbol{\theta}$:

$$q(\boldsymbol{\theta}) = \prod_{j=1}^d \phi_{0, \tau^2}(\theta_j) = (2\pi\tau^2)^{-\frac{d}{2}} \exp\left(-\frac{1}{2\tau^2} \sum_{j=1}^d \theta_j^2\right)$$

- MAP:

$$\begin{aligned}\hat{\boldsymbol{\theta}}^{\text{MAP}} &= \arg \min_{\boldsymbol{\theta}} (-\log p(y | \boldsymbol{\theta}, \mathbf{x}) - \log q(\boldsymbol{\theta})) \\ &= \arg \min_{\boldsymbol{\theta}} \left(-\log p(y | \boldsymbol{\theta}, \mathbf{x}) + \frac{d}{2} \log(2\pi\tau^2) + \frac{1}{2\tau^2} \sum_{j=1}^d \theta_j^2 \right) \\ &= \arg \min_{\boldsymbol{\theta}} \left(-\log p(y | \boldsymbol{\theta}, \mathbf{x}) + \frac{1}{2\tau^2} \|\boldsymbol{\theta}\|_2^2 \right)\end{aligned}$$

- We see how the inverse variance (precision) $1/\tau^2$ controls shrinkage

